Public EV Charging Planning Report

Site-readiness and right-sizing model results

Future City Fellowship Phase 2 prototype

George Mason University

Report title	Public EV Charging Planning Report
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Purpose	Screen candidate public EV charging sites, right-size starter installations, and identify review needs.
Use boundary	Planning support only; not engineering, permitting, procurement, or construction approval.

Planning-support artifact only. These outputs support early screening, prioritization, and discussion. They do not replace final engineering design, utility coordination, permitting, procurement, ADA review, fire/AHJ review, stormwater review, construction approval, site survey, or stakeholder confirmation.

Executive Summary

This report summarizes the latest run of the public EV charging site-readiness and right-sizing model. It is intended to compare candidate sites, identify stronger starter opportunities, flag review needs, and separate near-term installation from future-ready planning.

12	2	5	15	0	38
CANDIDATE SITES	TIER 1 STARTERS	DECISION CATEGORIES	TABLEAU EXPORTS	BLOCKING ERRORS	REVIEW ITEMS

Key findings

- FFX002 - City of Fairfax Regional Library is the strongest current planning signal, with readiness score 58.8.

- Tier 1 starter candidates in this run are: FFX002 - City of Fairfax Regional Library; FFX004 - Fairfax County Government Center.

- Scenario testing identifies future-ready capacity gaps for: PWC002 under combined_stress; PWC008 under combined_stress.

- All results remain planning-screen indicators and require facility, utility, ADA, fire/AHJ, permitting, procurement, and stakeholder confirmation before implementation.

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Current Site Ranking

The ranking below orders candidate sites by the current readiness score. The score is a screening signal only; it should guide review and discussion rather than determine final approval.

Rank	Site	Score	Tier	Planning recommendation
1	FFX002 - City of Fairfax Regional Library	58.8	Tier 1	Advance to site screening and stakeholder confirmation before design or procurement.
2	FFX004 - Fairfax County Government Center	54.4	Tier 1	Advance to site screening and stakeholder confirmation before design or procurement.
3	PWC006 - Bull Run Library	52.1	Tier 2	Keep as a promising candidate and improve demand, parking, operating-hours, and electrical assumptions.
4	FFX001 - Fairfax City Hall or municipal building	51.0	Tier 2	Keep as a promising candidate and improve demand, parking, operating-hours, and electrical assumptions.
5	PWC002 - Neabsco-Potomac Commuter Garage	50.3	High-review	Keep in the planning discussion, but require garage, fire/AHJ, access, and electrical review before advancement.
6	PWC007 - Central Library	50.2	Tier 2	Keep as a promising candidate and improve demand, parking, operating-hours, and electrical assumptions.
7	PWC004 - Chinn Park Regional Library	48.6	Tier 2	Keep as a promising candidate and improve demand, parking, operating-hours, and electrical assumptions.
8	FFX003 - Pohick Regional Library	46.8	Tier 2	Keep as a promising candidate and improve demand, parking, operating-hours, and electrical assumptions.
9	PWC008 - Potomac Library	40.1	Tier 3	Monitor as a lower-priority candidate and revisit after stronger data is available.
10	PWC001 - PWC McCoart Government Center	39.8	Tier 3	Monitor as a lower-priority candidate and revisit after stronger data is available.
11	PWC005 - Haymarket Gainesville Library	36.9	Tier 3	Monitor as a lower-priority candidate and revisit after stronger data is available.
12	PWC003 - Prince William County Public Safety Training Center	29.6	Special-use	Do not advance as a general-public charging site until the public-access or fleet-only use case is confirmed.

Decision Buckets

Decision buckets translate the model outputs into planning categories that are easier to discuss with stakeholders.

Decision category	Meaning	Sites
Advance / strong starter candidates	These are the strongest current starter candidates in the model.	FFX002 - City of Fairfax Regional Library; FFX004 - Fairfax County Government Center
Good starter candidates with expandable design	These sites may support starter installations, but should be planned with expansion and verification in mind.	PWC006 - Bull Run Library; FFX001 - Fairfax City Hall or municipal building; PWC007 - Central Library; PWC004 - Chinn Park Regional Library; FFX003 - Pohick Regional Library
Resolve review issues before expansion	These sites may be useful, but review burden or electrical constraints should be addressed before major expansion.	PWC002 - Neabsco-Potomac Commuter Garage; PWC001 - PWC McCoart Government Center
Monitor / small pilot	These sites should be monitored or considered for smaller pilot-style planning rather than immediate expansion.	PWC008 - Potomac Library; PWC005 - Haymarket Gainesville Library
Do not advance for general-public buildout yet	A site should not advance as a general-public charging site unless public-access or fleet-only use-case assumptions change.	PWC003 - Prince William County Public Safety Training Center

Right-Sizing Recommendations

Right-sizing separates install-now ports from future-ready capacity. Install-now ports represent starter planning. Future-ready capacity indicates what the site should be prepared to accommodate if demand grows.

Site	Install now	Future-ready	Strategy / review note
FFX002 - City of Fairfax Regional Library	2	8	Simulation supports this as a conservative planning recommendation.
FFX004 - Fairfax County Government Center	3	8	Simulation supports this as a conservative planning recommendation.
PWC006 - Bull Run Library	2	4	Simulation supports this as a conservative planning recommendation.
FFX001 - Fairfax City Hall or municipal building	2	4	Simulation supports this as a conservative planning recommendation.
PWC002 - Neabsco-Potomac Commuter Garage	2	3	Simulation supports this as a conservative planning recommendation.
PWC007 - Central Library	2	4	Simulation supports this as a conservative planning recommendation.
PWC004 - Chinn Park Regional Library	2	4	Simulation supports this as a conservative planning recommendation.
FFX003 - Pohick Regional Library	2	4	Simulation supports this as a conservative planning recommendation.
PWC008 - Potomac Library	2	2	Simulation supports the right-sizing recommendation under tested assumptions.
PWC001 - PWC McCoart Government Center	2	3	Simulation supports this as a conservative planning recommendation.
PWC005 - Haymarket Gainesville Library	2	2	Simulation supports the right-sizing recommendation under tested assumptions.
PWC003 - Prince William County Public Safety Training Center	2	2	Simulation supports the right-sizing recommendation under tested assumptions.

Review Burden

Review flags are not failures. They identify inputs or site conditions that need confirmation before final design, procurement, permitting, or implementation decisions.

Error	Warning	Review	Info
0	0	38	12

Recurring confirmation fields

- electrical service rating
- spare electrical capacity
- emergency disconnect visibility
- visitor/employee counts
- operating hours
- parking count
- likely charger placement
- garage/fire/AHJ/access review

Scenario Sensitivity

Scenario testing checks whether future-ready capacity remains adequate under alternate planning assumptions, including higher growth, longer charging sessions, shorter charging windows, peakier demand, and combined stress.

Scenario	Maximum future-ready gap	Sites with a gap
Base assumptions	0	None
High growth	0	None
Long sessions	0	None
Short daily window	0	None
Peakier demand	0	None
Combined stress	1	PWC002: gap 1; PWC008: gap 1

Planning Interpretation

Use the report as a starting point for prioritization. Strong starter candidates should move into facility, utility, ADA, fire/AHJ, and stakeholder confirmation. Sites with higher review burden should resolve implementation questions before major expansion. Sites with lower public-priority or special-use signals should not be treated as general-public buildout candidates unless the public-access or fleet-only use case changes.

Planning-support disclaimer

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